

Highlights

The Evolution of Plane Rotations: From Classical Geometry to Geometric Algebra

Tibin Thomas, M J Vedan, Jessin Francis

- Reviews classical and modern representations of rotations across dimensions.
- Reinterprets rotations through invariant oriented planes rather than unique rotation axes.
- Represents oriented rotational planes by bivectors and rotations by rotors.
- Provides a coordinate-free formulation applicable to arbitrary dimensions and metric signatures.

The Evolution of Plane Rotations: From Classical Geometry to Geometric Algebra

Tibin Thomas^{a,*,1}, M J Vedan^{b,2} and Jessin Francis^{c,3}

^aAssistant Professor, Department of Mathematics, St Berchmans College, Changanasserry, 686101, Kerala, India

^aResearch Scholar, Department of Computer Application, CUSAT, 682022, Kerala, India

^bFaculty of Science, Cochin University of Science and Technology, CUSAT, 682022, Kerala, India

^cLecturer, Department of Electronics Engineering, Government Polytechnic College, Vechoochira, 686511, Kerala, India

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ABSTRACT

This paper traces the historical and mathematical development of plane rotations, from early coordinate-based descriptions through complex numbers, rotation matrices, Euler angles, quaternions, dual quaternions, and Lie groups, to the framework of Geometric Algebra. Although these representations differ widely in form, each encodes the rotational plane only indirectly, through matrices, angular parameters, algebraic elements, or infinitesimal generators. We argue that this plane, rather than the more familiar rotation axis, is the natural object underlying rotational motion, a distinction that becomes essential beyond three dimensions, where a unique axis no longer exists. Using Geometric Algebra, we show how oriented plane elements, represented as bivectors, generate rotations directly through the rotor formalism, recovering classical rotation formulas from a small set of algebraic axioms. This construction extends naturally to arbitrary dimensions and, through a change in metric signature, to Lorentz boosts in spacetime. The result is a coordinate-free and dimension-independent language for rotational transformations, one that unifies several classical frameworks within a single geometric picture.

1. Introduction

Rotation is one of the oldest concepts in geometry, yet how we represent it has kept evolving. From early coordinate descriptions of planar motion to the abstract machinery of Lie groups, each generation has developed new tools to describe rotations more compactly or more generally. Rotation matrices, Euler angles, quaternions, dual quaternions, and Lie theory all emerged from this search, and each remains in active use, from robotics and computer graphics to theoretical physics.

All these frameworks describe rotations in different ways. A matrix encodes a rotation through an array of numbers, Euler angles through a sequence of elementary rotations, quaternions through an algebraic element, Lie theory through an infinitesimal generator. In every case, what one writes down is a stand-in for something more basic, the plane in which the rotation actually takes place.

This matters most once we leave three-dimensional space. In three dimensions every rotation has an axis, and this picture is so familiar that it is easy to mistake for a universal feature of rotation rather than a coincidence of three-dimensional geometry. In four or more dimensions a plane no longer has a unique perpendicular direction, and the notion of an axis disappears. What survives, in every dimension, is the rotational plane itself, together with an angle. This suggests that the plane, not the axis, is the true object of study.

This paper traces the path to that conclusion and shows how Geometric Algebra gives the rotational plane a direct algebraic form. Section 2 reviews the historical development of plane rotations. Section 3 examines how rotation changes beyond three dimensions. Section 4 surveys rotation matrices, Euler angles, quaternions, dual quaternions, and Lie groups, and notes how each leaves the plane implicit. Section 5 develops a coordinate-free viewpoint centered on oriented plane elements. Section 6 introduces Geometric Algebra, showing how bivectors and rotors recover the familiar rotation formulas from a small set of axioms. Section 7 applies this formulation to curved spaces and spacetime, where the same construction generates both rotations and Lorentz boosts. Section 8 compares the frameworks and looks

*Tibin Thomas

 tbnmaths@gmail.com (T. Thomas); mjvedan@gmail.com (M.J. Vedan); jessinfrancis@gmail.com (J.F.)

 (T. Thomas); (M.J. Vedan); (J.F.)

ORCID(s):

ahead, and Section 9 concludes.

In this paper we are trying to show that the history of rotation theory can be read as a gradual recognition that rotations are transformations of planes rather than rotations about axes, and that Geometric Algebra offers the clearest setting in which to make this precise.

2. Origins and Historical Development of Plane Rotations

The concept of rotation has its origins in classical Euclidean geometry, where rigid motions preserving distances and angles were studied through geometric constructions. Early investigations treated rotations primarily as geometric transformations rather than algebraic objects. The introduction of analytic geometry allowed rotational transformations to be described using coordinates. Within Cartesian coordinates, a planar rotation through angle θ may be written as

$$\begin{aligned}x' &= x \cos \theta - y \sin \theta \\y' &= x \sin \theta + y \cos \theta.\end{aligned}$$

These equations constitute one of the earliest algebraic descriptions of planar rotations and can be expressed compactly in matrix form. The introduction of coordinate geometry marked a turning point in the study of rotations. Geometric constructions could now be expressed algebraically, opening the way for more compact and general mathematical representations.

A major conceptual advance occurred through complex numbers. In 1797, Caspar Wessel recognized that complex numbers could be interpreted as ordered pairs of real numbers, thereby representing both magnitude and direction in the complex plane, that is complex numbers can be interpreted as a vector in $\mathbb{C}$. In 1806 Jean Robert Argand independently published the same idea. Because Argand's work was published in a more widely read French journal, his name became synonymous with the representation. Carl Friedrich Gauss further popularized the concept, which is why it is also occasionally referred to as the Gaussian plane. By representing points in the Euclidean plane as complex numbers, planar rotations could be expressed through multiplication. This interpretation revealed that multiplication by $z' = e^{i\theta} z$ produces a rotation through the angle θ , transforming planar rotations into simple algebraic operations.

Complex number representations possess several advantages. It is a compact representation and has a natural geometric interpretation. However, extending complex arithmetic beyond planar geometry proved difficult. Suppose one attempts to represent directions in space using $x + iy + jz$, where i and j denote independent imaginary directions. A natural requirement is that multiplication should preserve geometric meaning and produce closed algebraic operations. Although imposing $ij = -ji$ improves certain algebraic properties, multiplication of such triplets fails to remain closed. Hamilton spent more than a decade searching for a three-dimensional analogue of complex numbers before discovering that a consistent division algebra necessarily requires four dimensions. This led to the invention of quaternions in 1843.

Quaternion algebra introduced four components $q = a + bi + cj + dk$ together with multiplication rules $i^2 = j^2 = k^2 = -1$ and $ij = k$, $jk = i$, $ki = j$ providing an algebraic framework capable of representing three dimensional rotations. Each stage attempted to preserve geometric intuition while extending rotational concepts beyond planar geometry. This naturally raises an important question. How should rotations be represented when dimensions continue to increase?

3. The Geometry of Rotations Beyond Three Dimensions

The study of rotations undergoes a fundamental conceptual change when one moves from two- and three-dimensional Euclidean spaces to higher dimensions. In the plane, every rotation is uniquely determined by a single angle, while in three-dimensional space rotations are commonly described by an axis and an angle. This geometric intuition has proved remarkably successful in classical mechanics, engineering and computer graphics. However, many of these familiar concepts no longer extend naturally to higher-dimensional spaces. This section examines why, and provides the motivation for the algebraic approaches discussed later.

3.1. From Planar Rotations to Spatial Rotations

In two-dimensional Euclidean space, every rotation takes place in the unique plane occupied by the entire space. Consequently, a rotation is completely determined by a single angle θ . The corresponding rotation matrix is

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which rotates every vector through the same angle while preserving distances and orientation. Since there is only one plane, the notion of a rotation axis does not arise.

The situation changes in three-dimensional Euclidean space. Here infinitely many planes exist, and every plane possesses a unique perpendicular direction. Consequently, a rotation is commonly described by an axis together with a rotation angle. This axis-angle representation has become one of the most familiar descriptions of three-dimensional rotations and forms the basis of numerous applications in mechanics, robotics and computer graphics. However, the rotation itself still occurs within a plane perpendicular to the axis. The axis therefore serves as a convenient geometric descriptor rather than the object undergoing the rotational transformation.

The transition to four-dimensional space introduces a fundamentally different geometric situation. A two-dimensional plane no longer possesses a unique normal direction, and consequently the notion of a unique rotation axis disappears. Instead, rotations are characterized by invariant rotational planes. Moreover, a four-dimensional rotation may consist of two independent rotations occurring simultaneously in two mutually orthogonal planes. Such transformations, known as *double rotations*, have no direct analogue in three-dimensional geometry and illustrate that higher-dimensional rotations cannot be fully understood through axis-based intuition.

This progression from planar rotations to spatial rotations demonstrates that increasing dimension changes not merely the number of coordinates but the underlying geometric nature of rotational transformations. The concept of a rotation axis, which is central in three dimensions, becomes a special feature of three-dimensional geometry rather than a universal property of rotations.

3.2. Rotational Planes in Higher Dimensions

The disappearance of a unique rotation axis in dimensions greater than three naturally shifts attention to the geometric object, the rotational plane, which remains invariant under rotation. In higher-dimensional Euclidean spaces, rotations are fundamentally transformations acting within two-dimensional subspaces, while vectors orthogonal to these planes remain unchanged or participate in independent rotations.

A *simple rotation* acts non-trivially in a single two-dimensional plane while leaving its orthogonal complement fixed. Such rotations generalize the familiar planar rotations of two-dimensional Euclidean geometry. In contrast, higher-dimensional spaces admit more general transformations in which several independent rotations occur simultaneously in mutually orthogonal planes. In four-dimensional Euclidean space, for example, *double rotations* have no analogue in three-dimensional geometry.

More generally, every proper orthogonal transformation in a finite-dimensional Euclidean space can be decomposed into rotations acting on mutually orthogonal invariant two-dimensional subspaces. This decomposition reveals that rotational planes, rather than rotation axes, constitute the fundamental geometric components of higher-dimensional rotations. The complexity of a rotation therefore depends not on the existence of an axis but on the number and arrangement of these invariant planes.

Theorem 1 (Plane Decomposition of Rotations). *Every proper orthogonal transformation on a finite-dimensional Euclidean space can be decomposed into a composition of rotations acting on mutually orthogonal invariant two-dimensional subspaces.*

This classical result follows from the real canonical form of orthogonal matrices and may be found in standard references on linear algebra and Lie groups.

3.3. Challenges in Higher-Dimensional Rotations

The geometric shift described above brings with it two practical difficulties.

- There is no longer a single axis to anchor a rotation.
- A general transformation may involve several independent plane rotations at once.

Both problems compound as dimension grows. The number of parameters required to describe a general rotation grows rapidly, and the corresponding algebraic representations become increasingly complicated. In an n -dimensional space, the number of independent rotational planes equal $\frac{n(n-1)}{2}$. For example when $n = 4$ it is six and for five dimensions ($n = 5$) it is ten. Coordinate-based approaches, while mathematically complete, often obscure the underlying geometric structure of the transformation. This motivates the search for formulations that retain the intrinsic geometry of rotations while remaining applicable in arbitrary dimensions.

From a computational perspective, higher-dimensional rotations arise in numerous areas including computer graphics, robotics, optimization, machine learning, molecular modeling and theoretical physics. Efficient algorithms must preserve orthogonality, numerical stability and computational efficiency while avoiding unnecessary redundancy in the representation of rotational transformations. These practical requirements have significantly influenced the development of modern rotation formalisms.

The geometric and computational challenges discussed above have motivated the development of increasingly sophisticated mathematical representations of rotations. Matrix methods, quaternions, Lie groups, Lie algebras and Geometric Algebra. Each address these challenges from different perspectives. The following section surveys these principal mathematical frameworks and examines their respective strengths and limitations in describing rotations in arbitrary dimensions.

4. Mathematical Generalizations of Rotations

The geometric challenges associated with higher-dimensional rotations have motivated the development of a variety of mathematical frameworks for representing rotational transformations. Each of these approaches was designed to address particular mathematical or computational requirements while preserving the essential properties of rotations, such as distance preservation and orientation.

Among these representations, rotation matrices provide the classical coordinate-based formulation and naturally extend to arbitrary dimensions. Euler angles offer an intuitive parameterization of three-dimensional rotations through successive elementary rotations, while quaternions furnish a compact algebraic representation that avoids several limitations of Euler angles. Dual quaternions further extend this framework to rigid body motions by incorporating translations together with rotations. At a more abstract level, Lie groups and Lie algebras provide a unified theoretical framework for studying continuous rotational symmetries.

Although these approaches differ in their mathematical formulation and areas of application, they share the common objective of representing rotations accurately and efficiently. The following subsections briefly review these principal methods, emphasizing their mathematical foundations, advantages and limitations, before introducing the plane-based viewpoint developed in the subsequent sections.

4.1. Rotation Matrices

The introduction of matrix algebra provided the first systematic framework for representing rotations in arbitrary-dimensional Euclidean spaces. Since rotations are linear transformations that preserve distances and angles, they can be described naturally by orthogonal matrices. Matrix representations remain the standard mathematical language for rotational transformations in linear algebra, numerical analysis, computer graphics and scientific computing.

An $n \times n$ real matrix R represents a rotation if it satisfies

$$R^T R = I, \quad \det(R) = 1,$$

where I denotes the identity matrix. The first condition ensures that inner products, lengths and angles are preserved, while the second excludes reflections. The collection of all such matrices forms the *special orthogonal group*

$$SO(n) = \{ R \in GL(n, \mathbb{R}) \mid R^T R = I, \det(R) = 1 \},$$

which plays a central role in the mathematical theory of rotations.

One of the principal advantages of matrix representations is their complete generality. The same formulation applies uniformly in every finite dimension, allowing rotations to be composed through matrix multiplication and readily incorporated into numerical algorithms. Matrix methods therefore constitute the foundation of many computational techniques in engineering, robotics, computer graphics and numerical linear algebra.

Several specialized matrix constructions have been developed for efficient numerical computation. For example, Givens rotations perform elementary plane rotations that eliminate selected matrix entries and are widely used in QR factorization and least-squares algorithms. Similarly, Householder transformations provide efficient orthogonal transformations for matrix reduction. These techniques illustrate the versatility of matrix-based representations in practical computations.

Despite these advantages, matrix representations describe rotations through their coordinate transformations rather than through the underlying geometric objects responsible for the rotation. In particular, the rotational planes are encoded implicitly within the matrix entries and are not represented explicitly. Consequently, while matrices provide a complete algebraic description of rotations in arbitrary dimensions, they offer comparatively limited geometric insight into the intrinsic structure of rotational transformations.

This observation motivated the development of alternative representations that seek either more compact parameterizations or a more direct geometric interpretation of rotations. One of the earliest and most influential of these approaches is the Euler angle representation discussed in the next subsection.

4.2. Euler Angles

One of the earliest systematic parameterizations of three-dimensional rotations was introduced by Leonhard Euler in the eighteenth century. Euler observed that an arbitrary spatial rotation can be represented as a sequence of three successive rotations about prescribed coordinate axes. This decomposition provided a practical method for describing the orientation of rigid bodies and laid the foundation for many applications in mechanics, astronomy and engineering.

Depending on the choice of coordinate axes and the order of successive rotations, several Euler angle conventions exist. A commonly used representation expresses a rotation as

$$R = R_z(\phi) R_y(\theta) R_x(\psi),$$

where ϕ , θ , and ψ denote the three Euler angles, and R_x , R_y , and R_z represent elementary rotations about the corresponding coordinate axes. Other conventions, such as the roll–pitch–yaw representation, use all three axes R_x , R_y , and R_z , are widely employed in robotics, aerospace engineering and computer graphics.

Euler angles possess several practical advantages. They provide an intuitive geometric description of orientation, require only three parameters, and are closely related to the physical motions of rotating bodies. Consequently, they remain one of the most widely used representations for describing rigid-body orientation in engineering applications.

Despite these advantages, Euler angle representations suffer from certain inherent limitations. The most well-known is *gimbal lock*, a singular configuration in which two rotational axes become aligned, causing the loss of one degree of rotational freedom. Furthermore, different Euler angle conventions may represent the same orientation, making interpolation and numerical implementation more complicated [1,2].

These limitations motivated the development of alternative parameterizations capable of representing three-dimensional rotations without singular configurations. Among the most successful of these are quaternion-based representations, which provide a compact algebraic formulation while avoiding gimbal lock.

4.3. Quaternions

The search for an algebraic representation of three-dimensional rotations beyond complex numbers led William Rowan Hamilton to the discovery of quaternions in 1843. Hamilton sought to extend the elegant properties of complex numbers, which describe planar rotations, to three-dimensional space. After many years of investigation, he recognized that such a generalization required four components rather than three, leading to the quaternion algebra $q = a + bi + cj + dk$, where $a, b, c, d \in \mathbb{R}$ and the imaginary units satisfy $i^2 = j^2 = k^2 = ijk = -1$. A quaternion may be written as $q = s + \mathbf{v}$, where s is the scalar part and $\mathbf{v}$ is the vector part. Rotations are represented by *unit quaternions*, satisfying $|q| = 1$, and a vector $\mathbf{x} \in \mathbb{R}^3$ is rotated according to $\mathbf{x}' = q \mathbf{x} q^{-1}$, where the vector is identified with the pure quaternion $\mathbf{x} = xi + yj + zk$.

Quaternion representations possess several important advantages over Euler angles. They avoid the phenomenon of gimbal lock, require only four parameters with a single normalization constraint, and permit efficient composition of rotations through quaternion multiplication. Furthermore, quaternion interpolation techniques, such as spherical linear interpolation (SLERP), have made quaternions the preferred representation for smooth rotational motion in computer graphics, robotics and spacecraft attitude control.

Despite these advantages, quaternion algebra is intrinsically associated with three-dimensional Euclidean space. Although quaternions provide an elegant and computationally efficient representation of spatial rotations, no analogous finite-dimensional division algebra exists that naturally extends quaternion rotations to arbitrary Euclidean dimensions. Consequently, while quaternions remain one of the most successful methods for representing three-dimensional rotations, they do not provide a universal framework for higher-dimensional rotational geometry [1,2].

This limitation motivated the development of more general mathematical frameworks. One important extension is provided by dual quaternions, which incorporate translational motion together with rotations and have become widely used in the representation of rigid-body transformations.

4.4. Dual Quaternions

While ordinary quaternions provide an efficient representation of three-dimensional rotations, many practical applications require the simultaneous description of rotations and translations. This need led to the development of *dual quaternions*, which extend quaternion algebra by incorporating dual numbers. Introduced by Clifford and later developed for rigid-body kinematics, dual quaternions have become an important mathematical tool in robotics, computer graphics and mechanical engineering.

A dual quaternion is expressed as $\hat{q} = q_r + \epsilon q_d$, where q_r and q_d are ordinary quaternions, ϵ is the dual unit satisfying $\epsilon^2 = 0$, $\epsilon \neq 0$, and quaternion multiplication extends naturally to the dual algebra. A unit dual quaternion represents a rigid-body displacement by combining a rotation with a translation into a single algebraic object.

One of the principal advantages of dual quaternions is their compact representation of rigid motions. They require fewer parameters than homogeneous transformation matrices, permit efficient composition through multiplication, and provide numerically stable interpolation of rigid-body motions. Consequently, dual quaternions have become widely used in robot kinematics, computer animation and motion planning [1,2,3].

Despite these advantages, dual quaternions remain fundamentally associated with rigid-body motions in three-dimensional Euclidean space. Their primary purpose is to unify rotations and translations rather than to provide a general framework for rotations in arbitrary dimensions. Consequently, they inherit many of the dimensional restrictions associated with quaternion algebra.

The search for mathematical frameworks capable of describing rotations uniformly in arbitrary dimensions therefore continued beyond quaternion-based methods. One of the most influential developments in this direction arose from the theory of continuous transformation groups, leading to the modern formulation of rotations through Lie groups and Lie algebras.

4.5. Lie Groups and Lie Algebras

The development of Lie groups and Lie algebras by Sophus Lie in the late nineteenth century provided a powerful mathematical framework for the study of continuous symmetries and transformation groups. Since rotations form continuous families of transformations, Lie theory offers a natural setting for analyzing their algebraic, geometric and differential properties. Today, Lie groups play a central role in differential geometry, theoretical physics, robotics and control theory.

The collection of all n -dimensional rotation matrices forms the *special orthogonal group* $SO(n) = \{R \in GL(n, \mathbb{R}) \mid R^T R = I, \det(R) = 1\}$, which is a Lie group under matrix multiplication. Associated with every Lie group there is a Lie algebra that describes its local or infinitesimal structure. For the rotation group $SO(n)$, the corresponding Lie algebra is $\mathfrak{so}(n) = \{A \in M_n(\mathbb{R}) \mid A^T = -A\}$, consisting of all skew-symmetric matrices. The relationship between the Lie algebra and the Lie group is established through the matrix exponential, $R = e^A$, where $A \in \mathfrak{so}(n)$. This correspondence enables finite rotations to be generated from infinitesimal rotational transformations and provides an elegant mathematical framework for studying continuous rotational motion [1,2,3].

One of the principal strengths of Lie theory is its generality. The same mathematical framework applies uniformly to rotations in arbitrary dimensions and extends naturally to more general transformation groups arising in differential geometry and mathematical physics. Furthermore, Lie groups provide the theoretical foundation for many modern algorithms in robotics, optimization, computer vision and quantum mechanics [1,2].

Despite its mathematical elegance, Lie theory represents rotations primarily through matrix groups and their associated infinitesimal generators. As a result, the underlying rotational planes remain implicit within the algebraic formalism. While Lie theory offers a profound understanding of the continuous structure of rotations, it does not explicitly represent the oriented planes that generate rotational transformations.

These observations motivate the search for mathematical frameworks in which the geometry of rotations is represented directly rather than indirectly through matrices or infinitesimal generators. This perspective naturally leads to the plane-based formulation developed in the following sections.

5. Coordinate-Free Rotations and Plane Elements

The mathematical frameworks discussed in the previous section describe rotations through matrices, angular parameters, quaternion algebra or continuous transformation groups. Although these representations differ in their mathematical formulation, they all describe the same underlying geometric phenomenon. This observation naturally raises a fundamental question: *Can rotations be understood independently of any particular coordinate system or algebraic representation?*

A coordinate-free viewpoint seeks to describe geometric transformations in terms of intrinsic geometric objects whose meaning does not depend on the choice of coordinates. From this perspective, coordinates merely provide numerical descriptions of geometric entities rather than defining the geometry itself. Consequently, the study of rotations shifts from coordinate transformations to the geometric structures responsible for rotational motion.

This section develops the coordinate-free interpretation of rotations by identifying oriented plane elements as the fundamental geometric objects underlying rotational transformations. These ideas provide the conceptual bridge between the classical mathematical representations discussed previously and the coordinate-independent formulation developed using Geometric Algebra in the next section.

5.1. Coordinate-Free Description of Rotations

Since the advent of analytic geometry, coordinate systems have provided one of the most effective tools for describing geometric objects and transformations. Rotations are commonly represented by matrices, angular parameters or algebraic quantities whose numerical values depend on the chosen coordinate system. Such representations have proved indispensable for computation and practical applications. Nevertheless, the geometric transformation itself is independent of the coordinates used to describe it.

A coordinate-free approach seeks to formulate geometry in terms of intrinsic geometric objects and relationships that remain invariant under changes of coordinates. From this viewpoint, coordinates serve merely as a convenient language for computation rather than as fundamental constituents of geometry. A change of coordinates alters only the numerical representation of a geometric object, while its geometric properties remain unchanged. Consequently, rotations should be understood as geometric transformations acting on the space itself rather than as manipulations of coordinate components.

This distinction becomes increasingly important in higher-dimensional spaces. While coordinate-based representations become progressively more complicated as the dimension increases, the underlying geometric nature of a rotation remains unchanged. Whether represented by a matrix, quaternion or other algebraic structure, every rotation preserves distances, angles and orientation. The essential geometric transformation is therefore independent of the particular coordinate system in which it is expressed.

The coordinate-free viewpoint shifts attention from numerical representations to the geometric objects responsible for rotational motion. Rather than asking how a rotation is represented in a given coordinate system, one asks what geometric entity generates the rotation itself. This perspective reveals that rotations are fundamentally associated with oriented two-dimensional subspaces, or *plane elements*, which remain meaningful regardless of the coordinate system employed.

Coordinate-free formulations have become increasingly important in modern geometry, differential geometry, theoretical physics and computational mathematics because they emphasize intrinsic geometric structure rather than coordinate-dependent descriptions. In the context of rotational geometry, this viewpoint provides the conceptual foundation for identifying plane elements as the fundamental objects of rotation, which forms the subject of the next subsection.

Proposition 1. *The geometric properties of a rotation are invariant under changes of coordinates; only its numerical representation changes.*

5.2. Plane Elements as the Fundamental Objects of Rotation

The coordinate-free viewpoint developed in the previous subsection shifts the emphasis from coordinate representations to the intrinsic geometry of rotations. From this perspective, the fundamental question is no longer how a rotation is represented, but rather which geometric object generates the rotational motion. The answer is provided by the concept of an *oriented plane element*.

In Euclidean geometry, every nontrivial rotation leaves a two-dimensional subspace invariant, while vectors orthogonal to this subspace remain unchanged or participate in independent rotations within other invariant planes.

Consequently, a rotation is fundamentally associated with an oriented plane rather than with an axis. The familiar axis of rotation in three-dimensional space is a consequence of the special geometry of $\mathbb{R}^3$, where every oriented plane possesses a unique orthogonal direction. In higher-dimensional spaces, however, this correspondence no longer exists, making the rotational plane the more fundamental geometric object.

A plane element may therefore be regarded as the geometric carrier of rotational motion. The orientation of the plane determines the sense of rotation, while the rotation angle specifies the magnitude of the transformation within that plane. Together, the plane and the angle completely characterize a simple rotation independently of the coordinate system used to describe it.

This viewpoint naturally extends to arbitrary dimensions. In an n -dimensional Euclidean space, a general rotation may be regarded as the composition of rotations acting in mutually orthogonal invariant planes. When the corresponding plane rotations act on orthogonal subspaces, they commute and may be performed independently. More general rotations arise through the composition of plane rotations whose associated planes are not mutually orthogonal, leading to richer rotational structures in higher dimensions.

The central role of plane elements unifies the geometric interpretation of rotations across all dimensions. In two dimensions there is a single rotational plane; in three dimensions the rotation axis is simply the normal to the rotational plane; and in four or more dimensions multiple independent rotational planes may coexist simultaneously. Thus, the apparent differences between rotations in various dimensions arise not from different geometric principles but from the changing relationships among their invariant planes.

The identification of oriented plane elements as the fundamental generators of rotations provides a coordinate-free geometric description that is independent of matrices, Euler angles, quaternions or other algebraic representations. The remaining challenge is to develop a mathematical framework in which these plane elements can be represented directly and manipulated algebraically. This challenge is addressed in the next section through the language of Geometric Algebra.

Theorem 2. *Every simple rotation in a finite-dimensional Euclidean space is uniquely determined by an oriented two-dimensional subspace and a rotation angle.*

This follows from the existence of an invariant two-dimensional subspace on which the rotation acts as an ordinary planar rotation, while vectors orthogonal to this subspace remain fixed.

5.3. From Plane Elements to Geometric Algebra

The coordinate-free interpretation of rotations developed in the preceding subsections establishes that the essential ingredients of a rotational transformation are an oriented plane and a rotation angle. This viewpoint is independent of any particular coordinate system or matrix representation and reflects the intrinsic geometry of the transformation. The remaining challenge is to identify a mathematical framework capable of representing these geometric objects directly while preserving their geometric meaning.

Most classical approaches describe rotations indirectly. Rotation matrices encode the transformation through arrays of numbers, Euler angles parameterize orientation by successive rotations, quaternions represent rotations using four-dimensional algebraic elements, and Lie groups characterize rotations through continuous transformation groups. Although these formulations differ significantly in their mathematical structure, they all describe the same underlying geometric phenomenon without treating the rotational plane itself as a primitive mathematical object.

A natural question therefore arises: *Can one develop an algebra in which the fundamental elements correspond directly to the geometric entities responsible for rotations?* Such a framework would combine the geometric clarity of coordinate-free reasoning with the computational advantages of algebraic manipulation. Instead of translating geometry into coordinates and then performing algebraic operations, the algebra itself would encode the geometry.

Geometric Algebra provides precisely such a framework. By extending the traditional vector space with new geometric elements representing oriented subspaces, it allows points, lines, planes and higher-dimensional objects to be treated within a single algebraic system. In particular, oriented plane elements become genuine algebraic objects that can generate and compose rotations in a natural and coordinate-independent manner.

This geometric interpretation unifies many of the ideas developed throughout the previous sections. The historical evolution from planar rotations to higher-dimensional rotational geometry, together with the limitations of purely coordinate-based representations, naturally leads to an algebra in which geometric objects themselves participate in algebraic operations. Consequently, Geometric Algebra is not introduced as an alternative computational technique, but as a mathematical language whose primitive objects reflect the intrinsic geometry of rotations.

The next section develops this formulation in detail by introducing the geometric product, bivectors and rotors, and demonstrating how rotations arise naturally from the algebra of oriented plane elements.

6. Geometric Algebra Formulation of Plane Rotations

Complex numbers provide an elegant algebraic description of rotations in the Euclidean plane, while quaternions extend this idea to three-dimensional space. However, neither framework naturally represents rotations in arbitrary dimensions. Since rotations fundamentally occur in planes, an algebra capable of representing vectors, planes and higher-dimensional subspaces within a single framework is desirable. Geometric Algebra provides such a framework by extending the vector space through the geometric product.

6.1. Why Geometric Algebra?

The historical development of rotational representations reveals a gradual search for mathematical frameworks capable of describing increasingly complex rotational motions. Complex numbers provide an elegant algebraic description of rotations in the Euclidean plane, while quaternions extend this idea to three-dimensional space. Matrix representations and Lie theoretic methods further generalize rotations to arbitrary dimensions and play an essential role in geometry, mechanics and computer science. However, each of these approaches possesses certain limitations.

Complex numbers are intrinsically two-dimensional and cannot represent rotations in arbitrary dimensions. Quaternions provide an efficient representation of three-dimensional orientations but do not extend naturally to spaces of dimension greater than three. Matrix representations are completely general but often obscure the underlying geometric meaning of a rotation, since the rotational plane is encoded implicitly within matrix entries rather than represented as a geometric object. Euler angle representations, although intuitive and widely used in engineering applications, are coordinate dependent and may suffer from singularities such as gimbal lock. Lie groups and Lie algebras provide a powerful theoretical framework for studying continuous rotational symmetries, but their geometric interpretation often requires additional mathematical machinery.

These observations point to what an ideal framework for rotations should look like. It should represent rotations directly in terms of the planes in which they occur, independently of any choice of coordinate system. It should extend naturally to arbitrary dimensions, and ideally unify Euclidean and non-Euclidean signatures within a single algebraic structure. Above all, it should preserve the geometric meaning of the rotation itself, rather than reducing it to a set of numbers.

Geometric Algebra satisfies these requirements by extending the classical vector algebra through a single associative product, called the *geometric product*. Unlike traditional approaches, Geometric Algebra represents not only vectors but also oriented planes, volumes and higher-dimensional geometric objects as algebraic entities called *multivectors*. In particular, bivectors naturally represent oriented planes and therefore provide the fundamental generators of rotations. Consequently, rotations are viewed not as motions about an axis but as transformations occurring within oriented planes, a viewpoint that extends naturally to spaces of arbitrary dimension.

The following subsections develop this formulation by introducing the geometric product, the emergence of bivectors, and the construction of rotors that generate rotations in a coordinate-free manner.

6.2. The Geometric Product

Classical vector algebra is based primarily on two binary operations, the scalar (dot) product and the vector (cross) product. While the dot product measures the projection of one vector onto another, the cross product is restricted to three-dimensional Euclidean space and therefore cannot serve as a general tool for describing rotations in arbitrary dimensions.

To overcome these limitations, Geometric Algebra introduces a single associative product, called the *geometric product*. For any two vectors a and b , it is defined as

$$ab = a \cdot b + a \wedge b, \quad (1)$$

where $a \cdot b$ denotes the symmetric (inner) product and $a \wedge b$ denotes the antisymmetric (outer) product. Since the inner product is symmetric,

$$a \cdot b = b \cdot a,$$

and the outer product is antisymmetric,

$$a \wedge b = -b \wedge a,$$

interchanging the vectors gives

$$ba = a \cdot b - a \wedge b.$$

Adding and subtracting the two equations immediately yields

$$a \cdot b = \frac{1}{2}(ab + ba), \quad (2)$$

and

$$a \wedge b = \frac{1}{2}(ab - ba). \quad (3)$$

Thus, the familiar inner and outer products arise naturally as the symmetric and antisymmetric parts of the geometric product. Unlike classical vector algebra, no additional operations are required ?,?,?.

The geometric interpretation of these two components is fundamentally different. The inner product is a scalar measuring the extent to which two vectors are parallel,

$$a \cdot b = |a||b| \cos \theta,$$

where θ is the angle between a and b . In contrast, the outer product measures the oriented parallelogram generated by the two vectors,

$$|a \wedge b| = |a||b| \sin \theta.$$

Consequently, the geometric product simultaneously contains information about both the relative orientation and the relative magnitude of two vectors. This unification is one of the principal advantages of Geometric Algebra and forms the algebraic foundation for representing rotations in arbitrary dimensions.

The outer product naturally introduces a new geometric object called a *bivector*, which represents an oriented plane element. Since rotations fundamentally occur in planes, bivectors become the natural generators of rotational transformations. This important observation forms the basis of the rotor formulation developed in the following subsections.

6.3. Bivectors as Oriented Plane Elements

The outer product introduced in the previous subsection naturally gives rise to a new geometric object called a *bivector*. For two vectors $a, b \in V$, the bivector generated by them is defined as $B = a \wedge b$. Unlike vectors, which represent directed line segments, bivectors represent oriented plane elements. Geometrically, the bivector $a \wedge b$ corresponds to the oriented parallelogram spanned by the vectors a and b . Its magnitude is given by

$$|B| = |a \wedge b| = |a||b| \sin \theta,$$

where θ denotes the angle between a and b . Thus, a bivector simultaneously encodes the plane determined by the vectors, the orientation of that plane, and the area of the corresponding parallelogram. An important property of the

outer product is its anti-symmetry, $a \wedge b = -b \wedge a$, which reflects the reversal of orientation of the plane. Consequently, $a \wedge a = 0$, showing that two parallel vectors do not generate a plane.

This geometric interpretation distinguishes Geometric Algebra from classical vector algebra. In three-dimensional vector analysis, the cross product $a \times b$ is often interpreted as representing the plane containing a and b . In reality, however, the cross product is an axial vector perpendicular to that plane and exists only in three-dimensional Euclidean space. It therefore represents the plane only indirectly through the Hodge duality.

By contrast, the bivector $B = a \wedge b$ lies in the plane itself and provides a direct algebraic representation of the rotational plane. This interpretation remains valid in every dimension and for arbitrary metric signatures. Consequently, bivectors provide a natural language for describing rotations in higher-dimensional spaces where a unique rotation axis generally does not exist.

This observation is fundamental to the present work. Since every rotation takes place within a plane, it is more natural to regard the rotational plane as the primary geometric object rather than an associated rotation axis. Geometric Algebra realizes this viewpoint by identifying bivectors as the generators of rotational transformations.

In Euclidean space, let $\{e_1, e_2, \dots, e_n\}$ be an orthonormal basis satisfying

$$e_i^2 = 1, \quad e_i e_j = -e_j e_i, \quad i \neq j.$$

Then the simple bivector

$$B = e_1 e_2 = e_1 \wedge e_2$$

represents the oriented coordinate plane spanned by e_1 and e_2 . Its square is

$$\begin{aligned} B^2 &= (e_1 e_2)(e_1 e_2) \\ &= e_1 e_2 e_1 e_2 \\ &= -e_1^2 e_2^2 \\ &= -1. \end{aligned}$$

. This property is of fundamental importance because it is precisely analogous to the imaginary unit of complex numbers,

$$i^2 = -1.$$

Thus, a unit bivector plays the role of the imaginary unit within its own rotational plane. Unlike the complex unit i , however, bivectors may represent arbitrary planes in spaces of any dimension. This observation forms the mathematical foundation of rotor theory, developed in the next subsection.

Remark 1. *The interpretation of bivectors as oriented planes provides a unified geometric description of rotations that is independent of the dimension of the underlying space. This viewpoint contrasts with the traditional axis-based interpretation, which is unique only in three-dimensional Euclidean space.*

6.4. Rotations Generated by Bivectors

The previous subsection established that bivectors provide natural algebraic representations of oriented planes. Since every rotation occurs within a plane, it is natural to expect that rotations should be generated by bivectors rather than by vectors or coordinate axes.

Let $B = e_1 e_2$ be a unit bivector representing the oriented plane spanned by the orthonormal basis vectors e_1 and e_2 . As shown previously, $B^2 = -1$. This property is analogous to the imaginary unit of complex analysis. Consequently, the exponential of a bivector may be defined through its power series,

$$e^{-B\theta/2} = \sum_{k=0}^{\infty} \frac{(-B\frac{\theta}{2})^k}{k!}. \quad (4)$$

Separating the even and odd powers and using the identity $B^2 = -1$, we obtain

$$e^{-B\frac{\theta}{2}} = 1 - \frac{B\theta}{2} + \frac{1}{2!} \left(-\frac{\theta^2}{4} \right) - \frac{1}{3!} B \left(-\frac{\theta^3}{8} \right) + \dots \quad (5)$$

$$= \cos \frac{\theta}{2} - B \sin \frac{\theta}{2}. \quad (6)$$

The multi vector

$$R = e^{-B\frac{\theta}{2}} = \cos \frac{\theta}{2} - B \sin \frac{\theta}{2} \quad (7)$$

is called a *rotor*. Since

$$R\tilde{R} = 1,$$

where $\tilde{R}$ denotes the reverse of R , its inverse is simply

$$R^{-1} = \tilde{R} = \cos \frac{\theta}{2} + B \sin \frac{\theta}{2}.$$

A vector v is rotated by the sandwich product

$$v' = RvR^{-1}. \quad (8)$$

This expression preserves both the magnitude and the grade of the vector while rotating it within the plane represented by B .

To illustrate, consider the basis vector e_1 . Using Equation (8),

$$e'_1 = Re_1R^{-1}.$$

Expanding the products and using the relations

$$Be_1 = -e_2, \quad e_1B = e_2,$$

one obtains

$$e'_1 = e_1 \cos \theta + e_2 \sin \theta. \quad (9)$$

Similarly,

$$e'_2 = -e_1 \sin \theta + e_2 \cos \theta. \quad (10)$$

Thus, the familiar two-dimensional rotation formulas are recovered directly from the rotor formalism.

Unlike matrix representations, the rotor is independent of any particular coordinate system. More importantly, the rotational plane is represented explicitly by the bivector B . Consequently, the same construction extends naturally to arbitrary dimensions by replacing B with the bivector corresponding to the desired plane of rotation.

In higher-dimensional Euclidean spaces, independent rotations may occur simultaneously in different planes. For example, the bivectors

$$B_1 = e_1e_2, \quad B_2 = e_3e_4,$$

represent two disjoint rotational planes. Since these bivectors commute,

$$B_1B_2 = B_2B_1,$$

the combined rotor

$$R = e^{-B_1\frac{\alpha}{2}} e^{-B_2\frac{\beta}{2}}$$

generates simultaneous rotations through the angles α and β in the respective planes. This illustrates how Geometric Algebra naturally accommodates multiple plane rotations, a feature that becomes increasingly important in higher-dimensional geometry.

Theorem 3 (Rotor Representation of Plane Rotations). *Let B be a unit bivector satisfying $B^2 = -1$. Then the rotor*

$$R = e^{-B\frac{\theta}{2}}$$

generates a rotation through angle θ in the plane represented by B , and every vector transforms according to

$$v' = RvR^{-1}.$$

All the standard text in geometric algebra contain the proof of this theorem, so here we are not including the proof.

Example 1. *Consider the Euclidean plane with orthonormal basis $\{e_1, e_2\}$ and let*

$$B = e_1e_2, \quad R = e^{-B\frac{\theta}{2}} = \cos \frac{\theta}{2} - B \sin \frac{\theta}{2}.$$

Using the sandwich product,

$$v' = RvR^{-1},$$

the basis vectors transform as

$$Re_1R^{-1} = e_1 \cos \theta + e_2 \sin \theta,$$

and

$$Re_2R^{-1} = -e_1 \sin \theta + e_2 \cos \theta.$$

Hence, the rotor reproduces the classical plane rotation matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

demonstrating that the bivector generates the familiar Euclidean rotation in the e_1e_2 -plane.

6.5. Plane Rotations in Higher Dimensions

Unlike two- and three-dimensional Euclidean spaces, higher-dimensional spaces admit simultaneous rotations in more than one independent plane. Consequently, the notion of a single rotation axis is generally no longer sufficient to describe a rotational transformation. Instead, a rotation is characterized by one or more oriented planes together with their corresponding rotation angles.

Within Geometric Algebra, each rotational plane is represented by a bivector. Let

$$B_1 = e_1e_2, \quad B_2 = e_3e_4,$$

be two unit bivectors corresponding to two mutually orthogonal planes.

Since these bivectors are constructed from disjoint basis vectors, they satisfy

$$B_1B_2 = B_2B_1.$$

Hence, the corresponding rotors commute,

$$R_1R_2 = R_2R_1,$$

and may be combined to form the composite rotor

$$R = R_1R_2 = e^{-B_1\frac{\alpha}{2}}e^{-B_2\frac{\beta}{2}},$$

where α and β denote the rotation angles in the respective planes.

The transformation of a vector is again obtained through the sandwich product

$$v' = RvR^{-1}.$$

Thus, multiple independent plane rotations are represented within the same algebraic framework without introducing additional coordinate systems or separate rotation matrices.

More generally, an arbitrary rotation in an n -dimensional Euclidean space may be decomposed into rotations in mutually orthogonal two-dimensional planes. Geometric Algebra expresses each of these planes directly through bivectors, allowing a unified and coordinate-free description of higher-dimensional rotations.

7. Applications and Extensions of Plane Rotations

7.1. Rotations in Higher-Dimensional Euclidean Spaces

The study of rotations becomes considerably more intricate as the dimension of the underlying space increases. In two-dimensional Euclidean space, every rotation occurs within the unique coordinate plane, while in three dimensions a rotation is commonly described by an axis together with a rotation angle. However, this axis-based interpretation does not extend naturally beyond three dimensions.

In an n -dimensional Euclidean space, a rotation is fundamentally characterized by one or more invariant two-dimensional subspaces, each associated with an independent rotation angle. Consequently, higher-dimensional rotations are more naturally understood as collections of simultaneous plane rotations rather than rotations about a single axis. We have seen the classical matrix theory expresses this through orthogonal transformations belonging to the special orthogonal group $SO(n)$, whose elements may be decomposed into rotations in mutually orthogonal planes.

From the viewpoint of Geometric Algebra, these rotational planes are represented directly by bivectors. Each bivector specifies the orientation of a rotational plane, while the associated rotor generates the corresponding rotation. Multiple independent rotations are represented by the composition of rotors, providing a unified geometric description that extends naturally to arbitrary dimensions.

This plane-based interpretation not only simplifies the conceptual understanding of higher-dimensional rotations but also provides a common language for Euclidean geometry, computational geometry and theoretical physics.

7.2. Rotations in Spacetime

The concept of plane rotations extends naturally beyond Euclidean spaces to the four-dimensional spacetime of special relativity. In Minkowski spacetime, rotational transformations include not only ordinary spatial rotations but also Lorentz boosts, which mix spatial and temporal coordinates. Together, these transformations constitute the Lorentz group, the symmetry group of special relativity.

Within the framework of Geometric Algebra, both spatial rotations and Lorentz boosts are generated by bivectors. The distinction between them arises from the metric signature of the plane in which the transformation occurs. In Euclidean space, a unit bivector satisfies

$$B^2 = -1,$$

leading to the trigonometric rotor

$$R = e^{-B\frac{\theta}{2}} = \cos \frac{\theta}{2} - B \sin \frac{\theta}{2},$$

which generates a rotation through an angle θ .

In contrast, a timelike bivector in Minkowski spacetime satisfies $B^2 = 1$, and the corresponding exponential becomes

$$R = e^{-B\frac{\phi}{2}} = \cosh \frac{\phi}{2} - B \sinh \frac{\phi}{2},$$

where ϕ is the rapidity. This rotor generates a Lorentz boost in the plane determined by the timelike and spacelike basis vectors.

Thus, Euclidean rotations and Lorentz boosts are represented by the same algebraic construction. The only distinction lies in the square of the generating bivector, which determines whether the exponential expands into trigonometric or hyperbolic functions. This unified treatment is one of the distinguishing features of Geometric Algebra and illustrates the close relationship between classical rotational geometry and relativistic transformations $?, ?, ?$.

The plane-based interpretation remains valid in spacetime. A spatial rotation occurs in a spacelike plane, whereas a Lorentz boost occurs in a timelike plane. Consequently, bivectors provide a common geometric language for describing both types of transformations without introducing separate mathematical formalisms.

Beyond special relativity, the same local viewpoint extends naturally to Lorentzian manifolds used in general relativity. At each spacetime point, the tangent space is a four-dimensional Minkowski vector space in which local Lorentz transformations act. Rotors may therefore be interpreted as local transformations of orthonormal frames, while the global geometry is determined by the curvature of the spacetime manifold. This observation suggests that Geometric Algebra provides a unified language for studying rotational phenomena in both flat and curved spacetimes.

7.3. Rotations in Curved Spaces

The discussion so far has been restricted primarily to Euclidean spaces, where rotations are global isometries preserving distances and angles. In many areas of modern mathematics and physics, however, the underlying space is curved rather than flat. Examples include surfaces embedded in three-dimensional space, Riemannian manifolds, Lorentzian manifolds in general relativity, and configuration spaces arising in mechanics and robotics. In such settings, the notion of a global rotation is generally not well defined, since tangent spaces at different points cannot be identified directly.

Instead, rotations are defined locally within the tangent space at each point of the manifold. If M is an n -dimensional differentiable manifold, the tangent space $T_p M$ at a point $p \in M$ is an n -dimensional vector space in which the tools of linear algebra apply. Consequently, every local rotation is again a rotation in a two-dimensional plane contained in the tangent space $T_p M$.

This local viewpoint preserves the fundamental geometric principle established throughout this paper is, rotations are intrinsically plane transformations. The curvature of the manifold affects how tangent planes are related to one another, but it does not alter the local structure of rotational motion within each tangent space.

In differential geometry, vectors are transported from one tangent space to another by means of a connection. Parallel transport describes how a vector changes as it moves along a curve while remaining as *parallel* as possible with respect to the chosen connection. When a vector is transported around a closed loop on a curved manifold, it generally returns with a different orientation. This change in orientation reflects the curvature of the underlying space and provides a geometric interpretation of holonomy.

From the viewpoint of Geometric Algebra, local rotations may be represented by rotors acting within the tangent space, while bivectors continue to represent the corresponding rotational planes. Thus, the rotor formalism developed for Euclidean spaces extends naturally to local orthonormal frames on curved manifolds. Rather than replacing the classical framework of differential geometry, Geometric Algebra provides an alternative algebraic language in which vectors, planes and rotations are represented within a unified structure.

This local formulation is particularly useful in applications where rotational quantities evolve continuously along curves or surfaces, such as rigid body motion on curved manifolds, geometric mechanics, gauge theories and general relativity. In these settings, the plane-based interpretation of rotations remains valid even though the ambient space itself is curved.

Remark. The local representation of rotations by bivectors suggests that Geometric Algebra may provide a natural framework for studying rotational structures on manifolds, including vorticity fields, frame transport, and rotational invariants. A detailed investigation of these topics lies beyond the scope of the present paper but represents an interesting direction for future research.

7.4. Emerging Applications of Plane Rotations

The mathematical description of rotations has become increasingly important in modern science and engineering. As problems involving high-dimensional data, complex geometric structures and relativistic phenomena continue to emerge, there is a growing need for rotational frameworks that are both geometrically meaningful and computationally efficient. Since Geometric Algebra represents rotations directly through bivectors and rotors, it provides a unified language for a wide range of applications.

In robotics and rigid body dynamics, accurate representations of orientation and motion are essential for navigation, control and motion planning. Rotor-based formulations provide smooth interpolation of orientations while avoiding the singularities associated with Euler angle parameterizations. Similarly, in computer graphics, computer vision and animation, rotations are fundamental in object manipulation, camera motion and skeletal animation. Plane-based representations offer an intuitive geometric interpretation that naturally extends to higher-dimensional transformations.

Higher-dimensional rotations also arise in numerical linear algebra, optimization and data analysis. Orthogonal transformations such as Givens rotations and Householder transformations play an important role in matrix factorization, least-squares problems and eigenvalue algorithms. Viewing these transformations as plane rotations provides additional geometric insight and suggests natural connections with the bivector formalism of Geometric Algebra.

In theoretical physics, plane rotations form the foundation of classical mechanics, electromagnetism and relativity. The unified treatment of spatial rotations and Lorentz boosts through bivectors and rotors makes Geometric Algebra particularly attractive for spacetime physics. Similar ideas have found applications in spinor theory, gauge theory and quantum mechanics, where geometric transformations are central to the mathematical formulation.

The concepts developed in this paper are also relevant to emerging areas such as geometric computing [2], computational geometry [2], manifold learning and topological data analysis, where geometric structures evolve in high-dimensional spaces. Although many of these applications continue to employ matrix-based methods, the intrinsic representation of planes and rotations offered by Geometric Algebra provides an alternative viewpoint that may simplify both theoretical analysis and computational implementation.

These diverse applications illustrate that plane rotations are not merely classical geometric objects but constitute a unifying concept across several branches of mathematics, physics and engineering. The historical evolution of rotational representations discussed in this paper therefore reflects an ongoing search for mathematical frameworks capable of describing increasingly complex geometric transformations.

Remark. The increasing importance of rotational phenomena in higher dimensional geometry, continuum mechanics and relativistic fluid dynamics suggests that plane-based formulations may play an important role in future mathematical models. In particular, the representation of rotational structures by bivectors provides a natural foundation for studying vorticity, frame transport and rotational invariants in both Euclidean and Lorentzian settings.

8. Discussion and Future Perspectives

The historical development of rotational representations reflects a continuous effort to describe geometric transformations with increasing mathematical generality and computational effectiveness. Beginning with the geometric constructions of Euclid and the algebraic interpretation of planar rotations through complex numbers, successive developments such as rotation matrices, Euler angles, quaternions, dual quaternions and Lie theoretic methods were introduced to overcome the limitations of their predecessors. Each of these approaches has made significant contributions to mathematics, physics and engineering, while simultaneously revealing new challenges when extended to higher-dimensional spaces.

One of the central observations emerging from this study is that the geometric nature of rotations changes with dimension. In two dimensions, a rotation is uniquely determined by a single angle, whereas in three dimensions it is commonly interpreted as a rotation about an axis. However, in dimensions greater than three, the notion of a unique rotation axis is generally no longer applicable. Instead, rotations are more naturally characterized by invariant two-dimensional subspaces, or rotational planes. This shift from an axis-based interpretation to a plane-based interpretation provides a unifying perspective for understanding rotations in arbitrary dimensions.

From this viewpoint, Geometric Algebra does not replace the classical methods discussed in the preceding sections; rather, it provides a mathematical framework in which many of these approaches can be interpreted within a common geometric language. By representing rotational planes as bivectors and rotations as rotor transformations, Geometric Algebra preserves the geometric meaning of rotations while extending naturally to higher-dimensional spaces and different metric signatures.

The following discussion compares the principal rotational representations reviewed in this paper and highlights their respective strengths, limitations and domains of applicability.

8.1. Comparative Analysis of Rotation Representations

Over the past two centuries, several mathematical frameworks have been developed to represent rotational transformations. Although these methods ultimately describe the same geometric phenomenon, they differ significantly in their mathematical formulation, geometric interpretation, computational implementation and range of applicability. We summarize the principal characteristics of the most widely used rotational representations discussed in this paper as remark shown below.

Remark 2 (Summary of Classical Rotation Representations). *The various representations of rotations discussed in this paper should be viewed as complementary rather than competing approaches. Their principal characteristics may be summarized as follows.*

- **Complex numbers** provide an elegant algebraic description of rotations in the Euclidean plane but are inherently restricted to two dimensions.
- **Rotation matrices** offer a general representation in arbitrary dimensions and form the basis of many numerical algorithms, although the underlying geometric structure is not represented explicitly.

- **Euler angles** provide an intuitive parameterization of three-dimensional orientations and are widely used in engineering applications. However, they depend on the chosen coordinate system and may suffer from gimbal lock.
- **Quaternions** furnish a compact and numerically stable representation of three-dimensional rotations, avoiding the singularities associated with Euler angles, but they do not admit a direct generalization to arbitrary dimensions.
- **Lie groups and Lie algebras** provide the natural theoretical framework for continuous rotational symmetries through the groups $SO(n)$ and their corresponding Lie algebras $\mathfrak{so}(n)$. Their abstraction, however, often obscures the underlying geometric objects.
- **Geometric Algebra** represents rotations through rotors generated by bivectors, thereby describing rotations directly in terms of their rotational planes. This formulation extends naturally to arbitrary dimensions and different metric signatures while preserving the geometric interpretation.

The comparison demonstrates that no single representation is universally optimal. Complex numbers remain the natural language for planar rotations, while Euler angles and quaternions continue to play important roles in three-dimensional applications. Matrix methods remain indispensable for numerical computations and linear algebra, whereas Lie theoretic methods provide the natural setting for studying continuous symmetry groups.

8.2. Future Perspectives

Although the mathematical theory of rotations has reached a high degree of maturity, several important challenges remain in extending rotational methods to increasingly complex geometric settings. The historical development reviewed in this paper suggests that future advances are likely to arise from formulations that preserve geometric structure while remaining applicable to arbitrary dimensions and metric signatures.

One important direction concerns the development of computational algorithms based on Geometric Algebra. While matrix methods remain the standard tool in numerical linear algebra and scientific computing, rotor-based formulations offer an alternative geometric representation that may simplify the implementation of rotational transformations in higher-dimensional spaces. A systematic comparison of computational efficiency, numerical stability and memory requirements between matrix, quaternion and rotor-based methods remains an interesting topic for future investigation.

Another promising area is the study of rotations on curved manifolds. Since rotations are defined locally in tangent spaces, the bivector representation provides a natural description of local rotational planes. This viewpoint may contribute to the study of frame transport, holonomy and geometric structures arising in differential geometry, although many theoretical questions remain open.

The extension of plane-based rotational formulations to spacetime geometry also presents significant opportunities. The unified treatment of Euclidean rotations and Lorentz boosts through bivectors suggests possible applications in relativistic continuum mechanics, spacetime kinematics and fluid dynamics, where rotational quantities play a fundamental role. In particular, the geometric interpretation of relativistic vorticity and rotational invariants may benefit from formulations based on Geometric Algebra.

Emerging fields such as geometric computing, robotics, computer vision, manifold learning and high-dimensional data analysis continue to generate new problems involving rotational transformations. As these applications become increasingly geometric in nature, mathematical frameworks that represent rotations intrinsically rather than purely through coordinates are likely to become more important.

The present survey has emphasized the conceptual evolution of rotational representations from classical geometric constructions to modern algebraic formulations. Future research may continue this development by combining the geometric insight of plane-based representations with advances in computational mathematics, differential geometry and mathematical physics.

9. Conclusion

The mathematical representation of rotations has evolved significantly over the past two millennia, reflecting both the increasing complexity of geometric problems and the development of new algebraic frameworks. Beginning with the geometric constructions of Euclidean geometry, the study of rotations progressed through complex numbers,

rotation matrices, Euler angles, quaternions and Lie theoretic methods, each providing new insights while addressing the limitations of earlier approaches. This historical development demonstrates that no single representation is universally applicable; rather, each framework is best suited to particular mathematical and computational contexts.

A central observation emerging from this survey is the gradual transition from an axis-based interpretation of rotations to a plane-based interpretation. While the notion of a rotation axis provides an intuitive description in three-dimensional Euclidean space, it does not extend naturally to higher dimensions, where rotations are fundamentally associated with invariant two-dimensional subspaces. This shift in perspective reveals that rotational planes constitute the essential geometric objects underlying rotational transformations in arbitrary dimensions.

Within this context, Geometric Algebra provides a unified mathematical language in which rotational planes are represented explicitly by bivectors and rotations are generated by rotors. This formulation extends from ordinary Euclidean rotations to Lorentz boosts without any change in the underlying construction, the same bivector generates both, and only the sign of its square, $B^2 = 1$ or $B^2 = -1$, decides whether the result is a rotation or a boost. Rather than replacing classical methods, Geometric Algebra incorporates many of their fundamental ideas into a single coordinate-independent framework based on geometric objects.

The survey has also highlighted the continued importance of plane rotations in modern mathematics, physics and engineering. Applications ranging from robotics and computer graphics to differential geometry and relativistic physics demonstrate that the study of rotations remains an active and evolving field. In particular, the extension of plane-based rotational formulations to curved manifolds and spacetime geometries suggests promising directions for future research.

Finally, this survey emphasizes that the historical development of rotational theory is more than a succession of algebraic techniques; it reflects a progressive refinement in our geometric understanding of rotational motion. The transition from coordinate-based descriptions to intrinsic geometric formulations, culminating in the bivector and rotor formalism of Geometric Algebra, provides a coherent perspective on rotations across dimensions. It is hoped that this unified viewpoint will contribute to future developments in higher-dimensional geometry, computational mathematics and mathematical physics. The history of rotation theory is, in many respects, the history of recognizing that rotations are fundamentally transformations of planes rather than rotations about axes.